

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459584.69075 +/- 0.00052 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1515 +/- 0.0062
Transit depth (Rp/Rs)^2	2.29 +/- 0.19 [%]
Orbital Inclination (inc)	85.02 +/- 0.97
Airmass coefficient 1 (a1)	1.2817 +/- 0.009
Airmass coefficient 2 (a2)	-0.1322 +/- 0.0028
Scatter in the residuals of the lightcurve fit is	0.69 %
Best Comparison Star	None
Optimal Aperture	7.07
Optimal Annulus	17.67
Transit Duration (day)	0.0838 +/- 0.0036

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1515 ± 0.0062 vs 0.1301 (deviation 1.99σ)
Duration fit vs published	0.0838 d vs 0.0908 d (deviation 1.13σ)
Mid-time O–C	1.7 min
Depth SNR	14.1
Residual scatter	0.69 %

Light curve

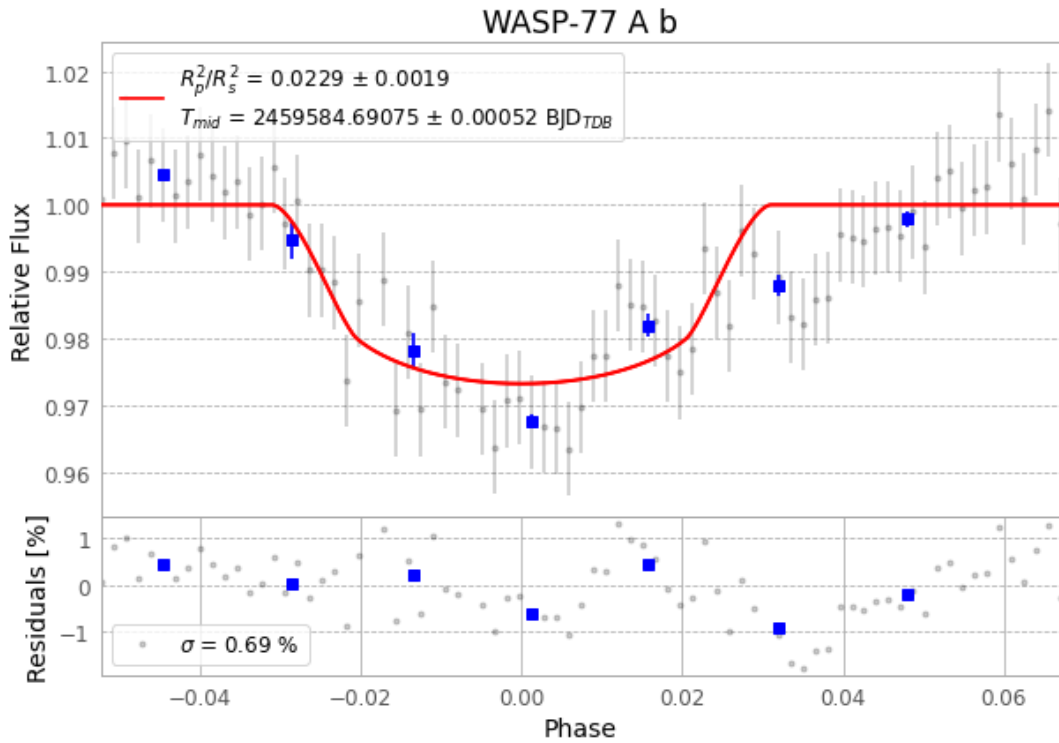


Figure 1 — FinalLightCurve_WASP-77 A b_2022-01-04.png (SHA-256 9982618284da3393...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [534, 191], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 0a5e0ee733a79a3a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

79 FITS frames (52 MB), WASP-77220105024812.FITS → WASP-77220105064212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-77-220105.log (093559aa794d...), FinalLightCurve_WASP-77 A b_2022-01-04.png (9982618284da...), FinalLightCurve_WASP-77 A b_2022-01-04.csv (d19d6b39559c...)

Compute resources

Wall time 145.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)